Project structure plan: BaSyx DPP API

Project 6: API for the Digital Product Passport (DPP) in the BaSyx Framework

Customer

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Definition of tasks

> **DIN EN 18222** “Digital product passport – Application programming interfaces (APIs) for product passport lifecycle management and searchability” describes a REST API that is to be implemented in the BaSyx framework as part of this task, both on the backend and frontend sides. The exact task can be found here.

Team 6

Name	e-Mail	Position
Nataliia Chubak	inf24271@lehre.dhbw-stuttgart.de	Project Manager
Luca Schmoll	inf24137@lehre.dhbw-stuttgart.de	Product Manager
Magnus Lörcher	inf24155@lehre.dhbw-stuttgart.de	Product Manager
Manuel Lutz	inf24224@lehre.dhbw-stuttgart.de	Test Manager
Noah Becker	inf24038@lehre.dhbw-stuttgart.de	System Architect
Fabian Steiß	inf24138@lehre.dhbw-stuttgart.de	Technical Writer
Felix Schulz	inf24075@lehre.dhbw-stuttgart.de	UI-Designer

Version Control

Version	Date	Author	Comment
1.0	12.10.2025	Nataliia Chubak	Created and added structure
2.0	14.10.2025	Nataliia Chubak	Added Milestones, Project Organisation and Gantt chart
3.0	15.10.2025	Nataliia Chubak	Risks analysis with suggestions from Luca Schmoll
4.0	21.10.2025	Nataliia Chubak	Deliverables

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1. Project Assignment

The goal of the project is to develop a complete, two-sided REST API solution for the Digital Product Passport (DPP). This solution is to be implemented in the BaSyx framework and must comply with the relevant DIN standards.

The outcome of this project lies in the establishment of standardized lifecycle management. The implementation of the solution is intended to improve the searchability of the DPP and increase overall service efficiency.

The main tasks include the phases of analysis, design, coding, documentation, and testing. The budget for this project is specified in detail in the business case (BC). The project will officially begin with the introductory lecture on September 19, 2025, and end with the final presentation and project submission in 2026 (exact end date to be determined).

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2. Project Context

During the project, we need to create and implement a REST API for a digital product passport (DPP). The project development involves the use of Eclipse BaSyx Framework, which includes frontend and backend. In addition, the project is also characterised by existing standard requirements and the lack of implementation of this standard in the Eclipse BaSyx platform.

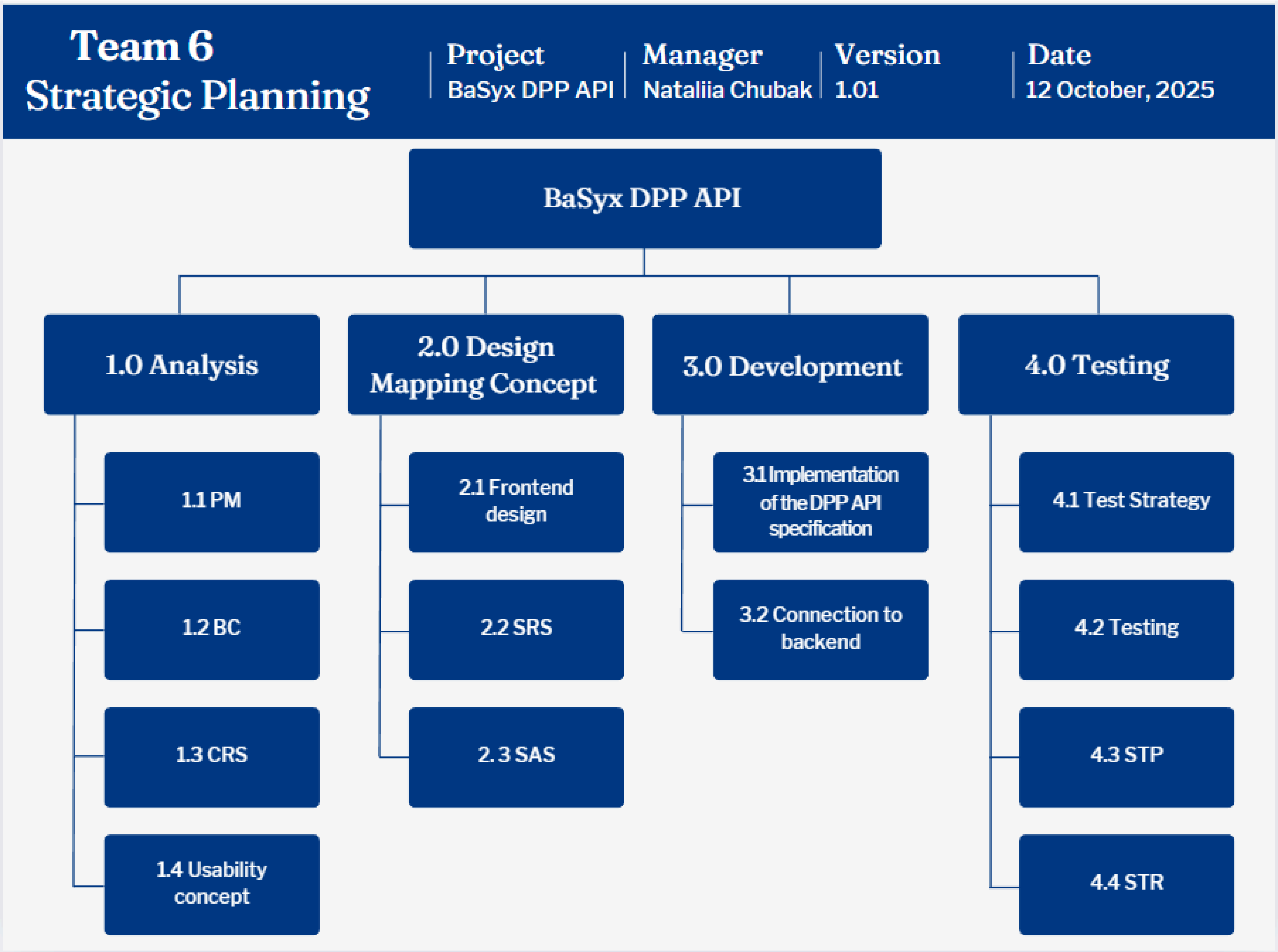
Time frame and phases:

- Platform basis: The BaSyx Framework is an open-source platform developed in 2023 for the development and use of digital twins in the context of Industry 4.0.
- Project phase: The current phase includes the complete implementation of the API, hosting on a demo server, and subsequent acceptance and integration into the open-source project.
- Post-project phase: After successful project completion, the BaSyx Framework will be expanded by the standards-compliant DPP API, which will significantly increase the interoperability of the platform.

3. Stakeholder Analysis

Stakeholder	Potential / Chance	Conflict / Risk	Actions
Customer	Satisfied with the project	Changing the requirements during the project	Regular communication between supplier and customer
Supplier	Development of the solutions that meets the requirements	Miscommunication, Time pressure	Regular meetings, Structured project leading
User	Uses the system	Needs more information or does not understand the provided documentation, Incorrect operation	Make documentation clear and easy to get. Create the usability concept and testing of catching errors

4. Work breakdown structure (PSP)



Figur.1 Work breakdown structure (PSP)

5. Milestones and deliverables

Phase I: Analysis

Nr	Milestone name	Week	Responsible person	Deliverable
M1	Project-Kickoff I	1	Whole team	1. Project kick-off protocol (protocol of the first meeting) 2. Task distribution matrix (roles and initial responsibilities)
M2	Analysing the requirements	2–4	Nataliia Chubak, Magnus Lörcher	1. Detailed stakeholder analysis 2. Business case (BC) 3. Customer Requirements Specification (CRS)
M3	Project plan	5–6	Nataliia Chubak	1. Project structure plan (initial rough schedule and scoping) 2. Initial product backlog (prioritised CRS requirements)

Phase II: Design

Nr	Milestone name	Week	Responsible person	Deliverable
M4	Repository-Setup & Issue-Management	7	Fabian Steiß, Manuel Lutz	1. GitHub repository (basic structure) 2. Standard templates (README.md, PR templates) 3. Issue tracker setup (CRS, BC, SRS, SAS, STP, protocols)
M5	Black-Box-Design (WHAT)	8–9	Noah Becker, Luca Schmoll, Felix Schulz, Manuel Lutz	1. Software Requirements Specification (SRS) 2. Software Architecture Specification (SAS) 3. Mockups/API design sketches, Usability/DX concept
M6	White-Box-Design (HOW)	9–10	Whole team	1. Definition of modules and assignment as work packages
M7	Final preparation	11	Whole team, Manuel Lutz	1. Final project plan update (parallelised tasks, dependencies) 2. PowerPoint presentation (first version)
M8	Semester-Review	21.11.25	Whole team	1. Presentation – results of analysis & design phase, final repository

Phase III: Development

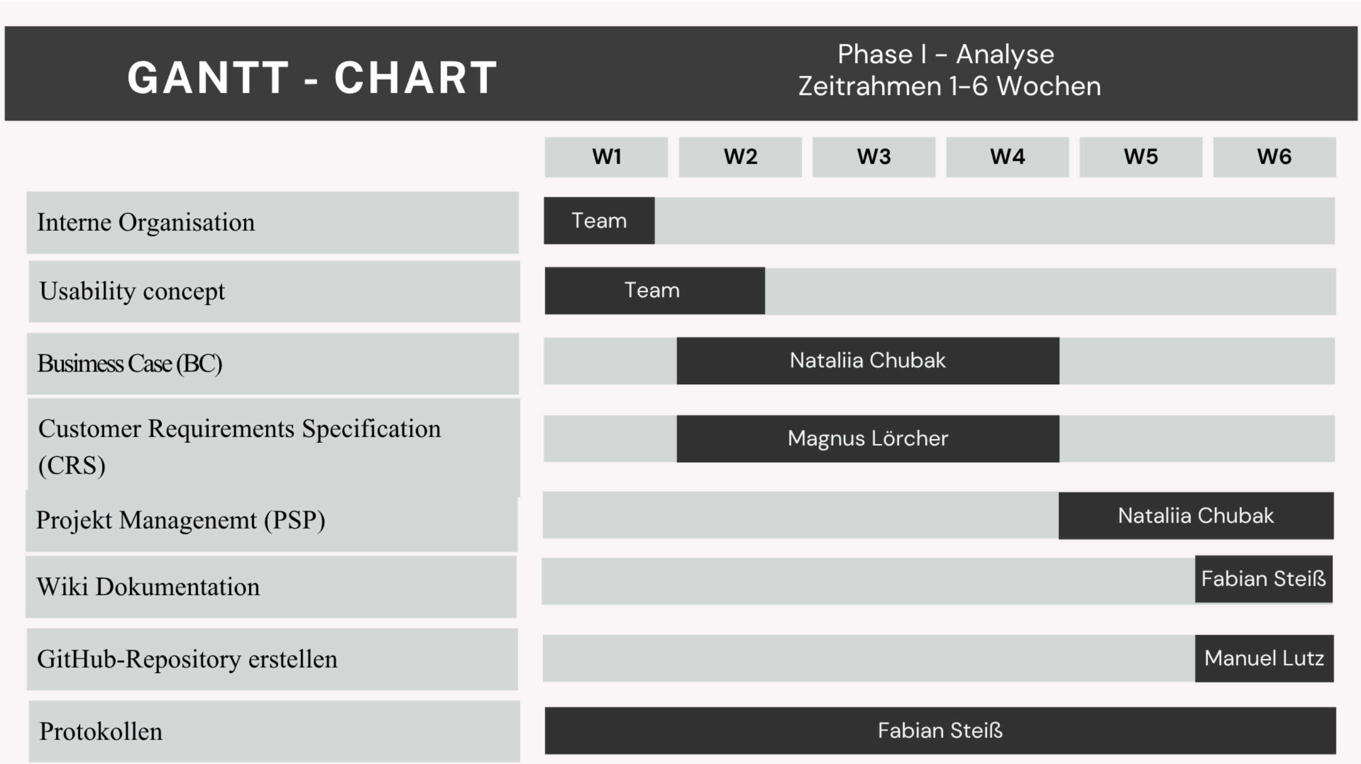
Nr	Milestone name	Week	Responsible person	Deliverable
M9	Project-Kickoff II	12	Nataliia Chubak	1. Update Project structure plan, including Implementation and Testing 2. Create Gantt-Diagram for 4th semester 3. Create task distribution matrix (initial responsibilities)
M10	Backend implementation	13-14	Fabian Steiß, Luca Schmoll, Magnus Lörcher	Implementation of backend into an exiting backend 1. Develop Java spring 2. Implementation of the DPP API-Specification 3. Update the BaSyx Backend services 4. Develop Docker Container 5. Mock backend data for frontend development
M11	Frontend implementation	15-16	Noah Becker, Felix Schulz, Nataliia Chubak	1. Connect to Backend 2. Frontend implementation 3. Develop Editor 4. Develop of Quick-Actions 5.Dashboard-Functions 6.Develop of Search/Filter Tabs
M12	Demo Server	17	Noah Becker	Configuration and protection of a server accessible via the internet for demonstating functionality

Phase IV: Testing

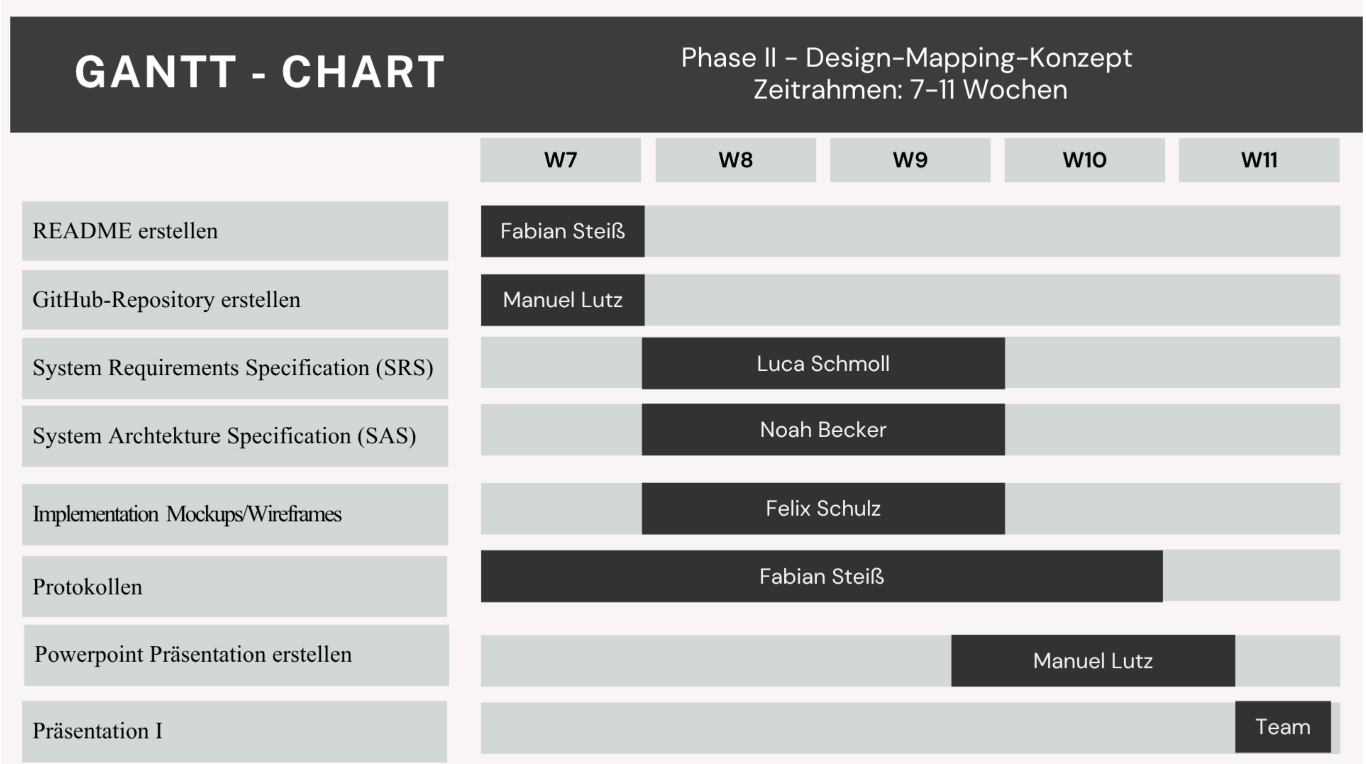
Nr	Milestone name	Week	Responsible person	Deliverable
M13	Functional testing	18	Manuel Lutz	1. Creating Issue-Tracker for Systemtest 2. Modular tests 3. Validation tests
M14	Integration and system tests	17	Manuel Lutz	1. End-to-end test 2. Interaction test
M15	Usability testing	19	Manuel Lutz	User Acceptance Test
M16	Documentation Review	20-21	Whole team	Preparation the documentation: 1. PM -> Nataliia Chubak 2. UserDocu -> Fabian Steiß 3. STR -> Manuel Lutz 4.STP -> Manuel Lutz 5. MOD (Web Interface) -> Felix Schulz 7. MOD

Nr	Milestone name	Week	Responsible person	Deliverable
				8. MOD 9. MOD 10. Meeting Minutes -> Nataliia Chubak
M17	Final Presentation	22.05.2026	Whole team	Final presentation – results of development & testing + final repository

6. Gantt chart 3rd Semester



Figur.2 Gantt chart 3rd Semester (Phase I - Analysis)



Figur.3 Gantt chart 3rd Semester (Phase II - Design)

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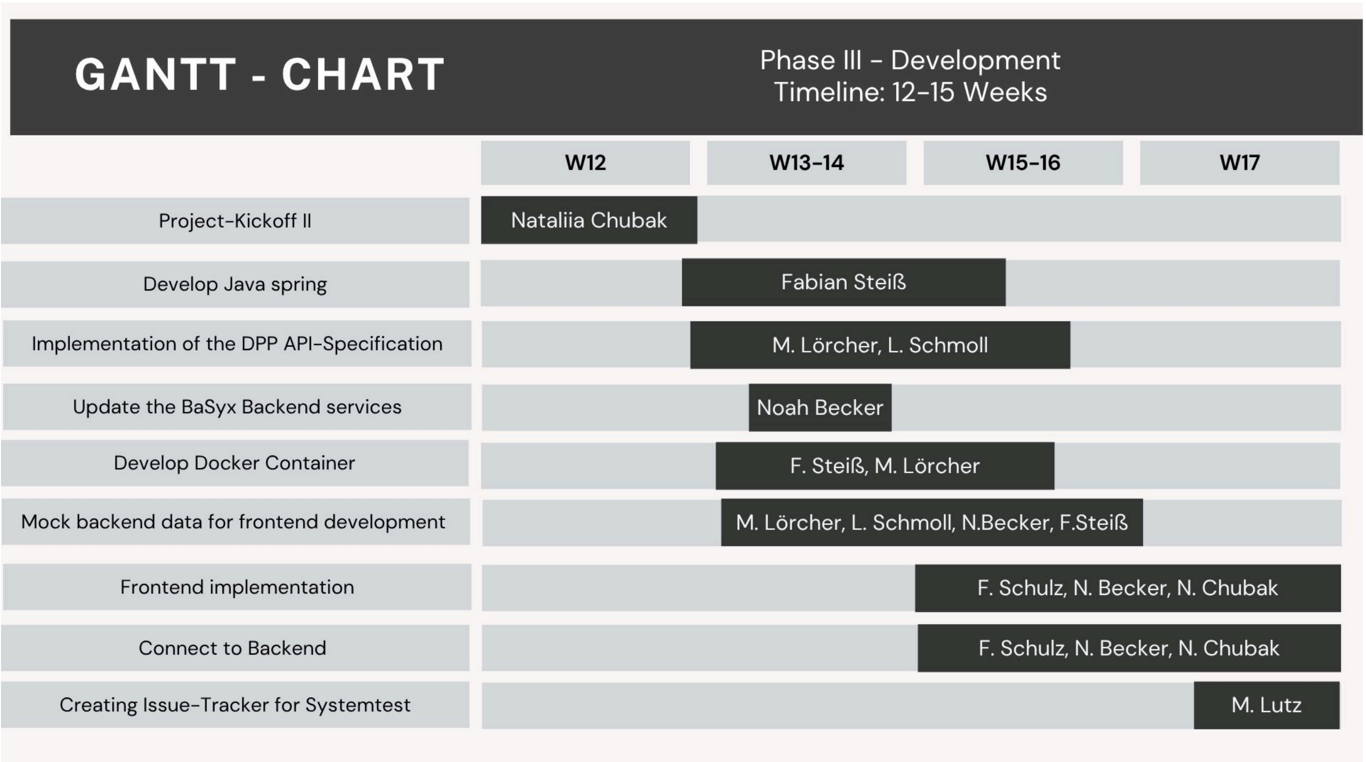
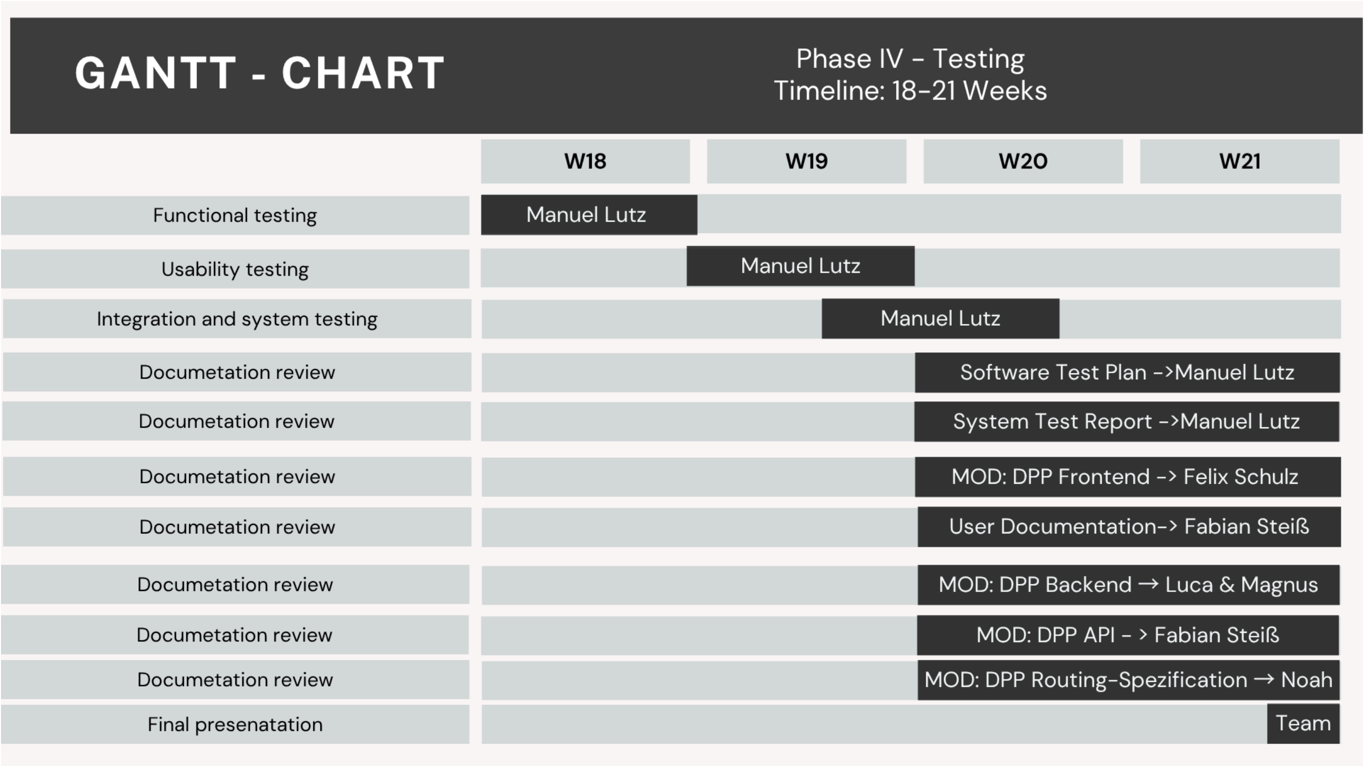
7. Gantt chart 4th Semester



Figur.5 Gantt chart 4th Semester (Phase III - Development)



Figur.5 Gantt chart 4th Semester (Phase IV - Testing)

List of tasks and responsible person

Person	**Task**
Nataliia Chubak	
Rolle: Project manager	- Planning & control
E-mail: inf24271@lehre.dhbw-stuttgart.de	- PSP (Project structure plan)
MatrikelNr: 6401719	- BC (Business Case)
	- Usability Concept
	- Coding (Frontend)
	- Presentation
	- Meetings Minutes (12-21 Weeks)
Magnus Lörcher	
Role: Product manager	- CRS (Customer Requirements Specification)
E-mail: inf24155@lehre.dhbw-stuttgart.de	- Market and demand analysis
MatrikelNr: 6699202	- Coding (Backend)
	- Usability concept
	- MOD: DPP Backend
	- PowerPoint II

Person	**Task**
	- Presentation
Luca Schmoll	
Role: Product manager	- SRS (System Requirements Specification)
E-mail: inf24137@lehre.dhbw-stuttgart.de	- Usability concept
MatrikelNr: 5919706	- Coding (Backend)
	- Market and demand analysis
	- MOD: DPP Backend
	- Presentation
	- PowerPoint II
Fabian Steiß	
Role: Technical writer	- Meetings Minutes (1-11 Weeks)
E-mail: inf24138@lehre.dhbw-stuttgart.de	- Usability concept
MatrikelNr: 5934347	- User Manual
	- MOD: DPP API
	- Readme
	- Wiki
	- Coding (Backend)
	- Presentation
Manuel Lutz	
Role: Test manager	- STP (Software Test Plan)
E-mail: inf24224@lehre.dhbw-stuttgart.de	- STR (System Test Report)
MatrikelNr: 9414567	- Test execution
	- Test planning
	- GitHub Repository
	- Coding (full stack)
	- Testing
	- Presentation
	- PowerPoint I
Noah Becker	
Role: System Architect	- SAS (Software Architecture Specification)

Person	**Task**
E-mail: inf24038@lehre.dhbw-stuttgart.de	- Backend implementation
MatrikelNr: 1871817	- Infrastructure setup
	- Usability concept
	- Coding (Full stack)
	- MOD: DPP Routing Spezification
	- Presentation
Felix Schulz	
Role: UI-Designer	- UI-implementation
E-mail: inf24075@lehre.dhbw-stuttgart.de	- Prototyping
MatrikelNr: 3954527	- BaSyx analysis
	- Executable
	- MOD: DPP Frontend
	- Coding (Frontend)
	- Presentation

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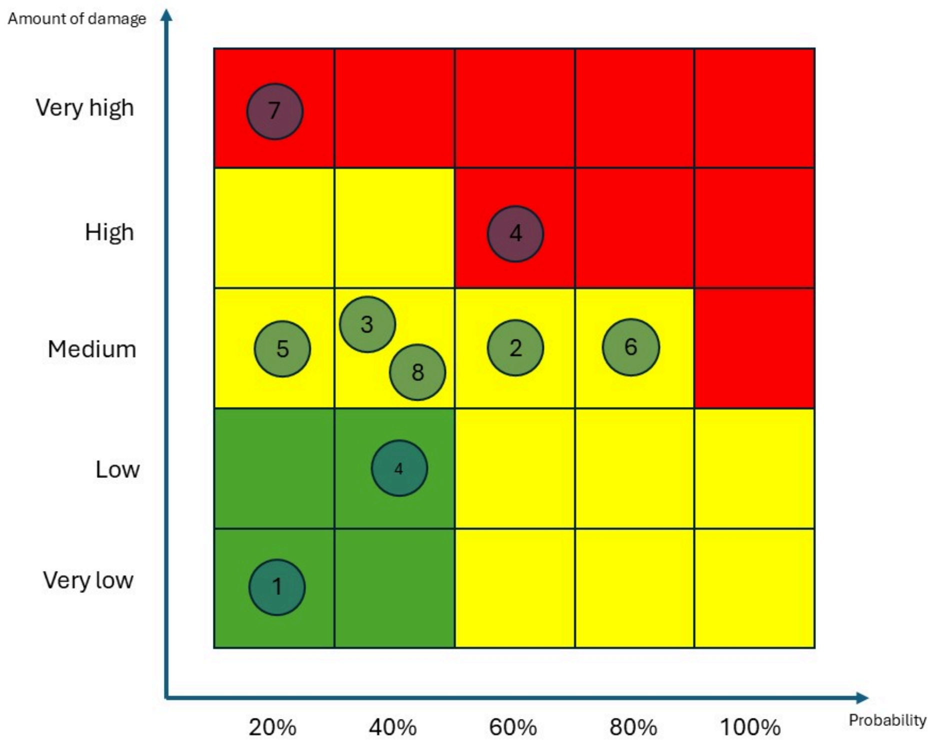
8. Risks

Nr	**Risk**	**Probability**	**Amount of damage**	**Effects**	**Measure**
1	Planning risk	20 %	Very low	The project might take longer than planned.	Creating a detailed project work plan and systematically monitoring of the progress.
2	Communication risk	40 %	Low	Insufficient communication between team members.	Holding regular meetings and use the GitHub and Jira services.
3	Miscommunication with client	40 %	Medium	Final product might not satisfy customer.	Presenting parts of the project during developing.
4	Technical risk	60 %	High	Technical complexity of BaSyx.	Evaluation and expansion of BaSyx teaching aids.

Nr	Risk	Probability	Amount of damage	Effects	Measure
5	Risk of ignoring risks	20 %	Medium	Insufficiently realistic assessment of scenarios by each member of the test team.	Creating a list of possible risks and possible solutions.
6	Budget risk	80 %	Medium	The budget for the project is being significantly exceeded.	Good planning and concentrating on main task.
7	Cyber attack	20 %	Very high	The servers are hacked; data is lost.	Encryption of the server.
8	Illness	40 %	Medium	Depending on the duration of the illness, several days/weeks.	In the event of long-term illness, the tasks assigned to the team member may be transferred to other team members.



Figur.6 Risk matrix

9. Soft- and Hardware requirements

Component	Technology
Backend	Java / Spring Boot (BaSyx SDK)
API Definition	OpenAPI 3.0 / Swagger
Frontend	React / TypeScript (BaSyx UI)
Data Model	Asset Administration Shell (AAS)
Infrastructure	Eclipse BaSyx Framework
Hosting	Traefik (Reverse Proxy) & Docker • Swagger OpenAPI Spezifikation • BaSyx WebUI Applikation
Documentation	Markdown, GitHub Wiki, Swagger UI

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10. Communication and reporting

- **Within the team:**

Meetings are every week. Meetings are held on the university premises. They are protocolled. Fabian Steiß is responsible for the protocols. At each meeting, every team member receives a task. In addition, every week, everyone reports on the progress of the task and any problems that may have arisen. The Teams platform is also used for communication, if clarification of a specific task is needed. The Jira platform is also used for more detailed control over the progress of tasks, which is also convenient for the team.

- **With the customer:**

All documents and implementation with testing processes are visible in the GitHub repository. The documents PM, BC, CRS, SAS, SRS, STR, STP, User Documentation and Meeting Minutes are in the PROJECT folder.

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11. Project deliveries

The project´s deadline is officially set on 22.05.2026.

Officially, the following tasks must be completed in the project:

- **GitHub:** The GitHub repository needs to be cleaned up, and its wiki has to be updated.
- **Documentation:** The complete set of documents must be delivered:
  - Project structure plan
  - Customer Requirement Specification CRS
  - Business Case BC
  - Software Requirement Specification SRS
  - Software Architecture Specification SAS
  - Protocols/Meeting Minutes Meeting Minutes
  - User Manual User Manual
  - Developer README (Setup, local development, backend & frontend) DevReadme
  - Mockups Mockups
  - MOD-Routing Routing

  - MOD-DPP-Frontend DPP-Frontend
  - MOD-DPP-BackendDPP-Backend
  - MOD-API API
  - System Test Report STR
  - Software Test Plan STP
- **Product presentation I&II** PowerPoint
- **Code: The source code and an executable version has to be published on GitHub Repository**